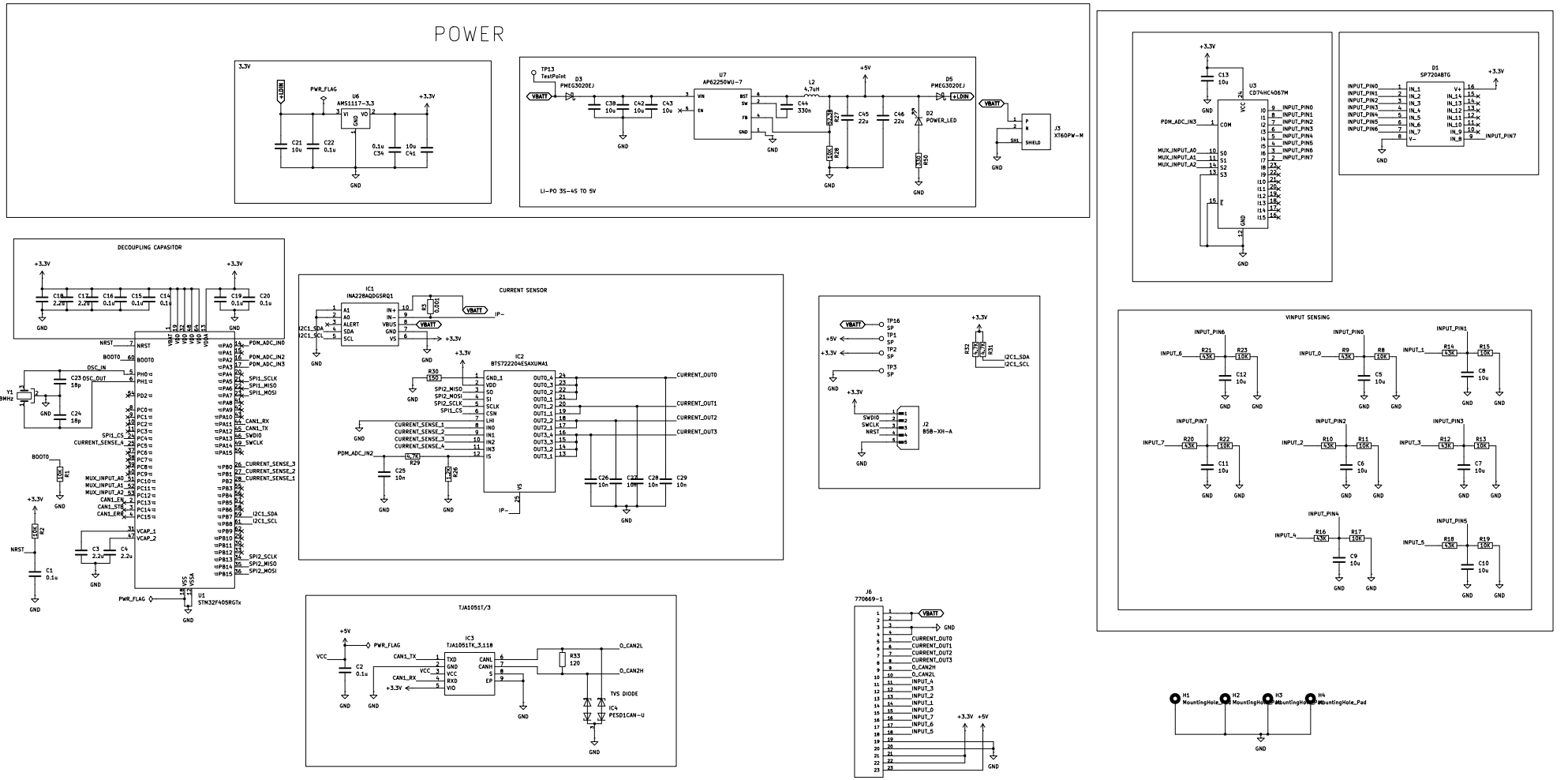

POWER
3.3V
+LDIN
PWR_FLAG
U6
AMS1117-3.3
VI
VO
GND
+3.3V
C21
10u
C22
0.1u
0.1u
C34
10u
C41
TP13
TestPoint
VBATT
D3
PMEG3020EJ
C38
10u
C42
10u
C43
10u
U7
AP62250WU-7
VIN
EN
BST
SW
FB
GND
L2
4.7uH
+5V
C44
330n
R27
100K
R28
10K
C45
22u
C46
22u
D5
PMEG3020EJ
+LDIN
D2
POWER_LED
R29
1K
R30
1K
VBATT
P
N
SHIELD
J3
XT60PW-M
LI-PO 3S-4S TO 5V
DECOUPLING CAPASITOR
C18
2.2n
C17
2.2n
C16
0.1u
C15
0.1u
C14
0.1u
C19
0.1u
C20
0.1u
NRST
BOOT0
OSC_IN
OSC_OUT
Y1
8MHz
C23
18p
C24
18p
R1
100K
R2
10K
C1
0.1u
C3
2.2u
C4
2.2u
VCAP_1
VCAP_2
PWR_FLAG
U1
STM32F405RGTx
PDM_ADC_IN0
PDM_ADC_IN2
PDM_ADC_IN3
SPI1_SCLK
SPI1_MISO
SPI1_MOSI
CAN1_RX
CAN1_TX
SWDIO
SWCLK
CURRENT_SENSE_3
CURRENT_SENSE_2
CURRENT_SENSE_1
CURRENT_SENSE_4
MUX_INPUT_A0
MUX_INPUT_A1
MUX_INPUT_A2
CAN1_EN
CAN1_STB
CAN1_ERR
I2C1_SDA
I2C1_SCL
SPI2_SCLK
SPI2_MISO
SPI2_MOSI
CURRENT SENSOR
IC1
INA228AQDGSRQ1
A1
A0
ALERT
SDA
SCL
IN+
IN-
VBUS
GND
VS
R3
100Ω
IP+
IP-
I2C1_SDA
I2C1_SCL
IC2
BTS722204ESAXUMA1
GND_1
VDD
SO
SI
SCLK
CSN
LHI
IN0
IN1
IN2
IN3
IS
OUT0_4
OUT0_3
OUT0_2
OUT0_1
OUT1_2
OUT1_1
OUT2_2
OUT2_1
OUT3_4
OUT3_3
OUT3_2
OUT3_1
R30
150
SPI2_MISO
SPI2_MOSI
SPI2_SCLK
SPI1_CS
CURRENT_SENSE_1
CURRENT_SENSE_2
CURRENT_SENSE_3
CURRENT_SENSE_4
PDM_ADC_IN2
4.7K
R29
C25
10n
1.2K
R26
C26
10n
C27
10n
C28
10n
C29
10n
CURRENT_OUT0
CURRENT_OUT1
CURRENT_OUT2
CURRENT_OUT3
TP16
SP
TP1
SP
TP2
SP
TP3
SP
R32
4.7K
R31
4.7K
I2C1_SDA
I2C1_SCL
J2
B5B-XH-A
SWDIO
SWCLK
NRST
TJA1051T/3
IC3
TJA1051TK_3.118
CAN1_TX
CAN1_RX
TXD
GND
VCC
RXD
VIO
CANL
CANH
S
EP
R33
120
O_CAN2L
O_CAN2H
TVS DIODE
IC4
PESD1CAN-U
J6
770669-1
INPUT_0
INPUT_1
INPUT_2
INPUT_3
INPUT_4
INPUT_5
INPUT_6
INPUT_7
U3
CD74HC4067M
PDM_ADC_IN3
MUX_INPUT_A0
MUX_INPUT_A1
MUX_INPUT_A2
COM
S0
S1
S2
S3
E
VCC
D1
SP720ABTG
INPUT_PIN0
INPUT_PIN1
INPUT_PIN2
INPUT_PIN3
INPUT_PIN4
INPUT_PIN5
INPUT_PIN6
INPUT_PIN7
VINPUT SENSING
R21
4.5K
R23
10K
C12
10u
R9
4.5K
R8
10K
C5
10u
R14
4.5K
R15
10K
C8
10u
R20
4.5K
R22
10K
C11
10u
R10
4.5K
R11
10K
C6
10u
R12
4.5K
R13
10K
C7
10u
R16
4.5K
R17
10K
C9
10u
R18
4.5K
R19
10K
C10
10u
H1
MountingHole_Pad
H2
MountingHole_Pad
H3
MountingHole_Pad
H4
MountingHole_Pad